Persistent Entity Handles and Local Reference Marks for a Frozen Robot Policy: A Preregistered Paired Simulation Study

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Abstract

Can finer spatial references help a frozen robot policy place an object more reliably without updating its weights or overriding its native action outputs? We study a simulator-assisted interface that binds stages to persistent entity handles and augments an existing placement marker with a parent-frame reference lattice and an object-center proxy. The original placement destination and native action chunks are preserved. A three-seed development comparison suggested a possible gain but provided only one discordant success pair. We then froze the method and registered 24 episodes on 12 locally unused seeds, counterbalancing arm order and requiring matched initial states and independently generated natural plans. The run was interrupted once by a host shutdown; one case that had recorded no environment step was re-executed after registration, and one case ended as an unknown infrastructure outcome. Under the registered accounting rule, 10 of 12 pairs are admissible: regions succeeded in 6/10 and point focus in 6/10, with 0 region-only and 0 point-only successes. The paired difference is 0.0 percentage points (conservative 95% interval $[-35.5, 35.5]$, exact paired $p = 1.0000$). Under the registered rule, the observed paired success difference is zero; the validation does not establish a region advantage or prove equivalence. Including the re-executed case, a disclosed deviation, adds one pair (seed 20261034) and gives regions 6/11 versus point focus 7/11 (-9.1 percentage points, $[-45.7, 32.7]$, $p = 1.0000$); this pair is the only source of that difference. Neither analysis supports a success benefit from the region presentation. Separately, 74 fixed-frame GPU generations exposed persistent visual verifier errors despite controller and prompt repairs, so the evaluated presentation excludes experimental alignment feedback. The study distinguishes persistent identity, spatial presentation, and reliable physical verification, and explicitly discloses simulator access and the limits of single-task evidence.

1 Introduction

A placement instruction names an object and destination, but a frozen visuomotor policy must still infer the local relation needed to execute it. An object-level marker may establish where a fixture is while conveying little about the neighborhood around its receiving site. This motivates a simple question: does an additional local spatial reference help an unchanged action policy use its existing manipulation competence?

We examine that question in a deliberately restricted setting. An execution wrapper maintains simulator-backed entity identities, binds naturally generated stages to those entities, and presents a fixed set of current image-space cues. During placement, the treatment adds neighboring reference points around the existing destination and a geometric-center proxy on the manipulated object. The complete placement goal remains present. Policy parameters remain fixed, and each native action chunk is executed without numerical modification. Figure 1 shows the saved policy inputs of both arms at one placement intervention.

The distinction between an implementation and an empirical improvement is central here. A stable handle can prevent a registry from forgetting an entity, but it does not make a vision-language model accurately recognize grasping or release. Similarly, a finer lattice does not constitute a geometric controller: the policy must interpret the marks, and a geometric center need not coincide with a cup's mouth or base. The treatment therefore evaluates a presentation package, rather than assuming that increasing the number of IDs necessarily increases physical precision.

Our development study suggested a possible benefit, but involved only three seeds and one discordant success pair. We consequently freeze the presentation and evaluate new seeds with unchanged model checkpoints, environment horizon, shared verifier, and action protocol. The primary comparison is regions against the shared point-focus controller, not against an unwrapped native policy. We separately report the failed attempt to use visual alignment feedback, rather than attributing its behavior to the region treatment. The prospective validation did not reproduce the development signal. Under the registered accounting rule, the two arms succeeded and failed together in every admissible pair, so no pair was discordant.

This manuscript contributes an auditable implementation of persistent entity-bound local reference prompting, a prospective paired test of its combined visual/text treatment,

and a failure analysis that separates structured-answer validity from visual correctness. Mark-based prompting, hierarchical robot control, and wrappers for frozen policies all have prior art; our claims are limited to the tested interface and evidence.

2 Related Work

Visual marks and spatial prompting. Set-of-Mark prompting exposes image regions to multimodal models through explicit visual marks at different granularities [1]. MOKA combines high-level task decomposition with marked keypoints and grids to infer affordances for manipulation [2]. PIVOT iteratively renders and refines candidate spatial proposals for a vision-language model to select [3]. These works preclude a broad novelty claim for marking objects, subdividing spatial references, or using hierarchical visual prompts. Our implementation instead evaluates a persistent simulator-backed binding and fixed parent-frame reference package at the input of an unchanged learned action policy. Unlike an explicit proposal-selection controller, it does not convert a selected lattice point into a motion command.

Geometric and hierarchical robot control. ReKep represents manipulation through functions over three-dimensional keypoints and uses hierarchical optimization to produce end-effector trajectories [4]. Our geometry supplies projected references but is not used to optimize a geometric control objective. Hi Robot couples high-level situated reasoning with low-level robot actions [5], establishing prior work on hierarchical instruction following. The narrow question here is whether the combined placement presentation helps a fixed policy under a shared execution protocol, rather than whether hierarchical decomposition itself is beneficial.

Wrappers around frozen policies. RoboHarness uses memory, retrieval, and a vision-language orchestrator to adapt frozen VLA policies [6]. Harness VLA combines frozen contact-rich policy primitives with analytic skills and execution memory [7]. Consequently, the general idea of a training-free execution wrapper is not unique to this work. Our intervention is restricted to entity-bound image/text presentation and the shared existing stage protocol; it adds neither a learned verifier nor analytic transport, placement, or release actions. The evidence should be interpreted as a controlled study of that restricted intervention, without comparing headline scores across incompatible tasks and access assumptions.

3 Method

We study an inference-time spatial prompting interface for an existing frozen robot policy. Let I_t denote the multi-camera observation, q_t the native proprioceptive input, and u_k the instruction for execution stage k. A persistent entity registry supplies the stage's object or region binding. A presentation function F produces marked images and a grounded instruction, which are consumed by the original policy:

$$A_t = \pi_\theta(F(I_{\le t}, u_k), q_{\le t}), \qquad \Delta\theta = 0. \tag{1}$$

The history in this expression is the native policy's current execution segment. The wrapper preserves the policy's action interface; it does not solve for an end-effector correction or overwrite the gripper command. "Training-free" refers to the absence of additional parameter updates in this method, rather than to the absence of pretraining or benchmark-specific training of the underlying checkpoint.

Simulator-backed persistent entities. The registry enumerates fixture and movable-object instances, geometric parts, and existing placement regions. Deterministic opaque handles are derived from simulator entity names and their parent relationships. A handle persists within an episode, including across stage changes and occlusions. Occlusion removes current projected geometry without expiring the entity. This is simulator-assisted identity retention, not a learned visual tracker or a demonstrated cross-episode re-identification capability. The natural planner receives the original visible candidate set; additional local reference points are not inserted into that set. The policy receives natural-language descriptions and colored visual cues rather than opaque identifier strings.

Parent-frame neighborhood. For every existing region with center c_t, we construct nine reference points indexed by $(i, j) \in \{-1, 0, 1\}^2$. Let R_t be the parent root-body rotation and let d_x, d_y be the planar extents of its root-local geometric axis-aligned bounding box (AABB), measured when the augmenter is constructed. The fixed spacing and reference points are

$$s = 0.15 \min(d_x, d_y), \qquad r_{ij,t} = c_t + R_t[is, js, 0]^\top. \tag{2}$$

The indices describe the parent's physical x/y frame, not image left/right. The construction is a point lattice, not a segmentation of the receiving surface into nine bounded cells. No point is labeled as the correct task answer. The original region center is retained at $(0, 0)$; each child handle is deterministic from its parent handle and index.

Object proxy and visibility. For every object we also compute the center of the union of its geometric AABBs in the root frame and transform that point by the current root pose. This proxy may include handles and other geometry. It is not a certified cup mouth, cup base, contact point, or control endpoint. Current camera transforms project the lattice and proxy into normalized image coordinates. New neighbor points require visible parent/region evidence and conservative ray visibility. The center child inherits the current original region projection and visibility, so the same original reference is not subjected to a different detector in the

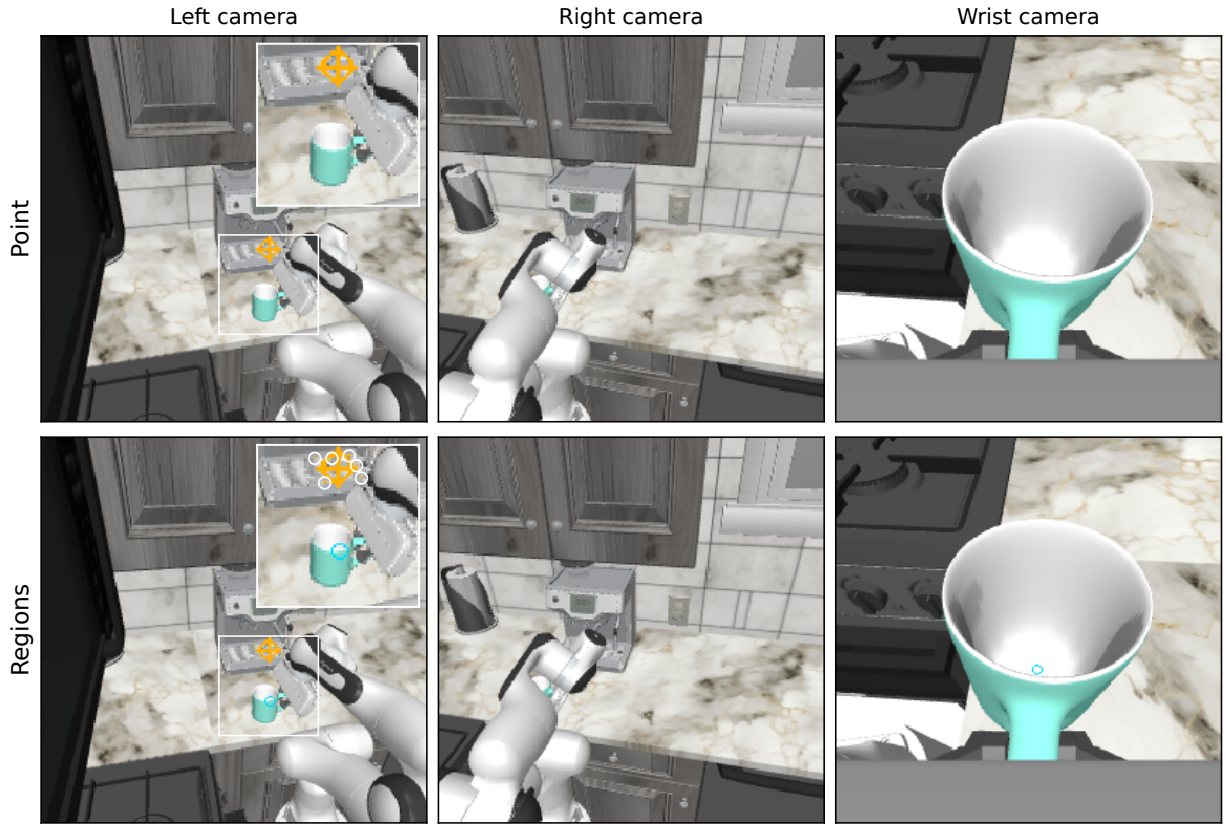


Figure 1: Actual saved policy inputs at the first placement intervention in development seed 20261025 (state 128). Both rows share the same raw observations. The point-focus crosshair (left camera) is retained in the region treatment. Relative to point focus, the region treatment changes 22 pixels in the left camera: six single-pixel orange neighbor points beside the crosshair and a cyan object-center proxy outline. The wrist camera receives only the proxy outline. No mark is drawn in the right camera at this step, so its two panels are identical. Insets show the boxed left-camera region with nearest-neighbor enlargement; white circles, added for this figure only, mark the six orange pixels. The fixed legend is not shown. Marks are displayed from the original 256-pixel inputs, without synthesizing a corrected scene. This is an input illustration from a failed development episode, not evidence of successful placement.

treatment. An object's proxy can be behind its own surface, but not behind a different occluding object. Missing or stale observations do not resurrect previous image coordinates.

Presentation treatments. The *point* arm uses the existing destination marker and related-entity cues within the shared execution controller. It is a point-focus control, not the unwrapped native policy. The *regions* arm retains that presentation and, only during active placement, adds visible orange neighboring points, a cyan object-proxy outline, and a fixed explanatory legend. At the original 256×256 resolution, neighbor glyphs are single pixels and the proxy is a radius-three outline. The center retains the original point marker. No connecting grid lines, opaque ID labels, error vectors, or numeric corrections are drawn. The legend explains the proxy limitation and asks the policy to use object shape and the camera views. Thus the treatment is a combined neighborhood/proxy/legend intervention; the present comparison does not isolate the causal contribution of ID granularity or of any one visual element.

Shared control protocol. Both arms use the same entity and geometry extraction rules, natural planning interface, frozen planner/verifier, visibility and identity checks, and native action interface. For a matched physical observation and stage context, the verifier presentation rule is identical; actual observations can diverge after the treatments produce different actions. The native policy is queried every 16 steps and returns the original 16×12 action chunk. Actions are executed without wrapper correction. A stage-instruction change resets the native execution segment according to the shared protocol. The expanded comparison excludes the experimental ALIGN/EXECUTE feedback treatment and its later prompt/parser repairs.

Simulator-access boundary. The method uses simulator entity membership, semantic asset-region definitions, segmentation, geometric bounds, poses, camera calibration, and

visibility queries. Only current two-dimensional projections and whitelisted descriptive metadata enter the model-facing context. Metric world geometry, contact diagnostics, and official task success are recorded separately and are not control inputs. Existing receiving sites are asset-authored semantic geometry and can coincide with the reference used by a benchmark's scorer; they are not inferred from RGB. Equal simulator access across arms therefore controls the treatment comparison but does not establish RGB-only grounding or real-world transfer.

Frozen models and scope. The evaluated policy is the frozen RoboCasa365 checkpoint of Xiaomi-Robotics-1 [8][1], with a frozen `Qwen3-VL-4B-Instruct` planner/verifier [9]. The registered evaluation is restricted to `CoffeeSetupMug` in the local RoboCasa environment [10, 11], with a 600-step horizon and left, right, and wrist RGB views. Exact checkpoint asset hashes, source hashes, simulator versions, and execution settings are retained through the frozen validation manifest and archived execution identities. This single-task evaluation is not a complete RoboCasa365 benchmark result.

4 Experimental Protocol

Development versus prospective validation. The previous development seeds are 20261025–20261027. Their outcomes and selected frames were inspected during implementation and verifier debugging, so they are not included in the new validation denominator. Before new execution, a local history audit selected and froze the 12 integer seeds 20261031–20261042. The claim of non-use is relative to the available local experiment records, not to unknown training data or deleted external records. Seeds are RNG identifiers rather than calendar dates. The registration consists of the validation manifest (digest `9414bd65...964eb9e4`) and an analysis plan marked frozen before execution. The plan was created at 16:10:21 UTC on 24 October 2026, five seconds before the launcher started; both were hashed locally before the first episode. Neither was deposited in an external registry, so their timing rests on local records retained in the release.

Fixed comparison. Each seed receives exactly one point episode and one regions episode, for 24 planned episodes. The two arms alternate which one runs first across the 12 pairs, yielding six pairs per order. The existing three-arm development controller is reused without feedback activation. No source, prompt, region spacing, marker size, action horizon, or decision threshold is tuned on validation outcomes. Completed task failures are not retried, seeds are not replaced, and all planned cases remain in the accounting.

[1] Hugging Face model ID: `XiaomiRobotics/Xiaomi-Robotics-1-RoboCasa365`.

The native policy adapter isolates Python, NumPy, and PyTorch CPU/CUDA random streams from the wrapper. It initializes the policy stream from the episode seed and retains it across proposals; the shared instruction-segment reset clears the native history and stream. These controls match the seeding protocol across arms without assuming that different prompted trajectories must produce identical actions.

Natural initial-state and plan matching. The first arm establishes a natural reference; the second resets from the same seed. Before any action in the second arm, the run checks the full initial signature, including the environment fingerprint, raw image and proprioceptive information, entity catalog, and shared geometry augmentation. It independently generates a natural plan and requires equality with the first arm's nonempty valid plan. A mismatch is retained as an invalid pair, without injecting the reference plan or forcing an initial state to pass. This design controls a common source of misleading comparisons, while restricting the paired estimand to admitted natural plans.

Execution and accounting. All model inference runs on the local RTX 5090 Laptop GPU (24,463 MiB); the policy and verifier remain CUDA-resident. CPU physics, preprocessing and logging are permitted. A single worker avoids concurrent model peaks. An eight-hour local watchdog includes startup and cleanup; it is a resource bound, not a shortened task horizon. Each complete episode retains the 600-step horizon. Official success ends an episode; reaching the horizon without success is a known task failure. Infrastructure interruptions, invalid pairing, and unattempted cases are reported separately as unknown, never silently converted to task failures or successes. The single exception, one zero-record case re-executed after a host shutdown, is described next; the primary analysis keeps that case unknown, as the registered rule requires.

Execution interruption and resume (post-registration deviation). The registered run was interrupted once by the host. After six episodes had finished (the complete pairs for seeds 20261031–20261033), a restart of the desktop process that owned the Linux subsystem stopped it. The launcher was terminated without cleanup. The seventh episode (seed 20261034, regions, first arm of its pair) had opened its trace, but no environment step, policy proposal or outcome had been recorded (zero trace records). The six recorded outcomes were kept, and none was rerun. A resume driver, written after registration, executed the 18 remaining cases (seeds 20261034–20261042) in a fresh directory, including that zero-step case. It reuses the registered manifest, site configuration, ports, checkpoints, frozen controller and verifier. It refuses to resume any case whose prior trace contains a step, and it requires the remaining cases to form whole pairs. The resume launch had its own eight-hour watchdog. Re-executing the zero-step case departs from the registered accounting rule: under that rule a case whose trace had been

opened is unknown, and the registered driver would not repeat it. We re-ran it because the trace shows the kill preceded the first environment step, so no step, model output or outcome had been observed. The primary analysis therefore follows the registered rule: the re-executed case stays unknown and the pair for seed 20261034 is excluded. The analysis that includes the re-executed case is reported separately as a deviation-dependent supplement. The final case (seed 20261042, point, second arm of its pair) had executed 544 of its 600 steps when one planner/verifier request exceeded the 180-second client timeout. Following the registered behavior, the worker stopped, and the launcher shut down all services without cleanup errors. The case is an unknown infrastructure timeout and was not repeated, so its pair is excluded from paired inference. The frozen analysis functions are applied through a wrapper that reads each case from exactly one of the two run directories. A combined evidence audit applies the frozen trace, array, renderer and CUDA checks to both directories. Because one case is unknown, its status is partial. It found every saved native action chunk mapped to a passing CUDA forward record (228 in the interrupted run, 424 in the resume, with no unmapped forward), and all 1,376 persisted planner/verifier generations passing their CUDA checks. The planner and policy service identities were identical across the two launches.

Outcomes and inference. The primary outcome is the environment's official success within the fixed horizon. For n valid complete pairs, let $s_i^R, s_i^P \in \{0,1\}$ be the region and point outcomes. We report

$$\widehat{\Delta} = \frac{1}{n}\sum_{i=1}^{n}(s_i^R - s_i^P) = \frac{b-c}{n}, \tag{3}$$

where b counts region-only successes and c point-only successes. We report the discordant counts and a two-sided exact McNemar test; no discordant pair gives $p = 1$. Per-arm success intervals and a paired effect interval accompany point estimates. All unknown denominators and bounds for unobserved paired outcomes are kept visible. With only 12 pairs, even a five-win/zero-loss pattern has exact two-sided $p = 0.0625$; this design can establish only large effects, and an inconclusive result does not prove equivalence.

The secondary horizontal diagnostic is the terminal object-root to selected receiving-site distance,

$$e_{xy} = \left\| x_{\mathrm{object},xy} - x_{\mathrm{site},xy} \right\|_2 . \tag{4}$$

It is computed offline from saved states, includes failures when available, and is not a cup-mouth alignment metric. We report coverage, per-seed values, and the median, rather than substituting it for official success. Contacts and benchmark thresholds are used only for offline diagnosis. Executed control steps are listed per seed in the ledger as descriptive context only. Neither steps nor wall-clock runtime were registered outcomes; the frozen analysis plan names only the terminal XY diagnostic as secondary.

5 Results

Development observations. Table 1 and Figure 2 include all three development seeds per arm. Regions improves one point failure to success and loses no successful point episode; two pairs tie. The exact paired test gives $p = 1$, which provides no statistical confirmation of benefit. The median distance decrease is descriptive and includes an episode whose distance worsened.

Table 1: Development evidence only (three episodes per arm). Terminal XY diagnostics include failures. Feedback is a separate, rejected experimental treatment.

Arm	Success	Median cm	Mean cm
Point	1/3	5.29	4.57
Regions	2/3	3.64	4.08
Feedback	1/3	6.99	13.11

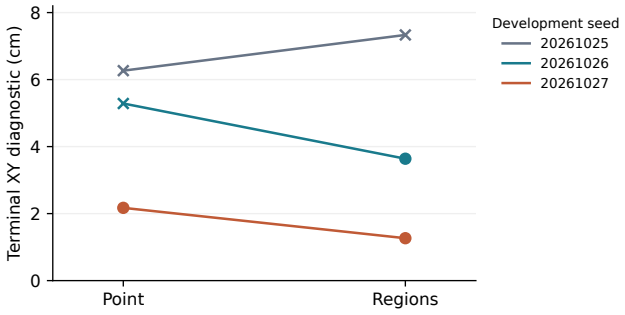


Figure 2: All three development pairs. Circles indicate official success and crosses failure. Lines connect the same seed; they do not represent learning curves. The region arm worsens one distance while improving two.

Prospective validation. The run was interrupted once by a host shutdown; one case that had recorded no environment step was re-executed after registration, and one case ended as an unknown infrastructure outcome. As executed, the fixed 24-case ledger (Table 3) contains 23 known task outcomes and 1 unknown case; under the registered rule the re-executed case is also unknown (22 known, 2 unknown). Under the registered accounting rule, 10 of 12 pairs are admissible: regions succeeded in 6/10 and point focus in 6/10, with 0 region-only and 0 point-only successes. The paired difference is 0.0 percentage points (conservative 95% interval $[-35.5, 35.5]$, exact paired $p = 1.0000$). The matched outcomes comprise 6 shared successes and 4 shared failures (Table 2). Under the registered rule, the observed paired success difference is zero; the validation does not establish a region advantage or prove equivalence.

Table 2: Prospective validation, separate from development. S/F/U are official successes, known task failures, and unknown cases across 12 planned episodes per arm. Paired success, Wilson intervals and the paired distance median use the identical admitted seed pairs of each analysis. Distances are median terminal XY diagnostics in centimeters, including failures, with n in parentheses; the all-known column also includes episodes whose partner is unknown and so compares different seed sets across arms. The registered-rule rows are primary; the deviation rows include the re-executed zero-step case (Section 4).

Analysis	Arm	S/F/U	Paired success	Wilson 95% (%)	Median cm, pairs	Median cm, all known
Registered rule	Point	7/4/1	6/10	[31.3, 83.2]	3.43 (10)	3.39 (11)
	Regions	6/5/1	6/10	[31.3, 83.2]	2.00 (10)	2.13 (11)
With deviation	Point	7/4/1	7/11	[35.4, 84.8]	3.39 (11)	3.39 (11)
	Regions	6/6/0	6/11	[28.0, 78.7]	2.13 (11)	4.76 (12)

Including the re-executed case. Including the re-executed case, a disclosed post-registration deviation, 11 of 12 pairs are admissible: regions succeeded in 6/11 and point focus in 7/11, with 0 region-only and 1 point-only success. The paired difference is -9.1 percentage points (conservative 95% interval $[-45.7, 32.7]$, exact paired $p = 1.0000$). The pair added by the deviation (seed 20261034: point success, regions failure) accounts for every difference between the two analyses. Neither analysis supports a success benefit from the region presentation.

The conservative effect intervals include zero. They are constructed from simultaneous exact marginal intervals, whereas the conditional paired test uses only discordant outcomes; both are reported without treating them as interchangeable.

For the 10 admissible pairs with both terminal distances (Figure 3), the median within-pair regions-minus-point difference was -0.31 cm and the mean was -0.57 cm. Including the re-executed case, the corresponding values over 11 pairs are -0.30 cm and 0.27 cm. Negative values favor regions. This secondary endpoint measures the natural selected site at each episode termination, not a common-time trajectory error or a semantic mouth-to-nozzle constraint. Official success remains the primary outcome.

Unknown cases are retained in the per-seed ledger and are excluded from paired task inference rather than relabeled as failures; complete case accounting does not mean all 24 robot trajectories completed. Bounding each excluded pair by its known arm and letting each unknown arm be either success or failure gives a worst-case envelope of $[-16.7, 0.0]$ percentage points over the 12 planned seeds under the registered rule, and $[-16.7, -8.3]$ including the re-executed case. These envelopes do not include sampling uncertainty and are not confidence intervals.

Treatment exposure (post hoc, descriptive). A separate audit of the retained traces and saved policy inputs checks whether the treatment was actually presented. It adds no test, conditions no rate on exposure, and excludes no case. All 24 episodes entered the Place stage; in all 12 pairs both arms entered it at the same step (steps 128–192). In all 12 pairs the native action chunks were identical before that step (8–12 chunks) and first differed at it. For seed 20261031, the raw camera image itself already differed between arms at step 55, before any treatment, while proprioception was identical; the raw images were identical again at step 192. Its source was not identified, but it precedes the treatment, and the action chunks stayed identical until the Place stage began at step 192. Region episodes recorded 5–29 rendered overlay receipts. Orange neighbor points were confirmed in the saved policy images in 9 of 12 region episodes. In the remaining 3 (seeds 20261035, 20261039, 20261041), no neighbor draw operation was recorded at any stored proposal boundary, although the legend was present. Point episodes recorded no hierarchy draw operations. Across all 24 episodes the frozen verifier committed the Place stage only once (seed 20261038, point arm); successes were ended by the environment's official check. Overlay receipts are stored only at proposal boundaries, so the number of frames on which the marks were visible is not known; these counts do not show that the policy read or used the marks.

6 Why We Exclude Alignment Feedback

An earlier experimental arm attempted to separate alignment and execution using a visual verifier. Its three development episodes did not produce any ALIGN-to-EXECUTE transition. Inspection also found a controller defect: the ALIGN sentence replaced the complete placement goal, instead of supplementing it. A later implementation preserves the original Place instruction and destination, resets confirmation streaks after invalid replies, and prevents duplicate frames from counting twice. These repairs passed contract tests but do not establish successful physical feedback.

Raw nulls versus parser failure. Across 161 hierarchical verification calls in the nine development episodes, 151 raw model responses left all five action facts (held, aligned, placed, released, clear) null. All 161 replies were valid JSON, and none hit the output token cap. In 48 replies, the entire nested answer object exactly copied the supplied null-filled template. These observations support template anchoring as a candidate explanation, but do not isolate a single

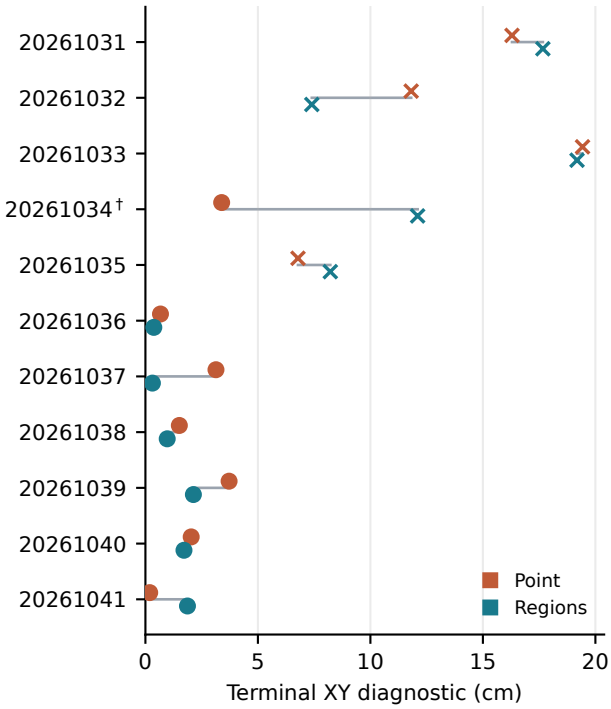


Figure 3: Terminal XY diagnostics for every prospective seed pair with both distances recorded. Each row is one seed; circles denote official success and crosses known failure, and the line joins the two arms of the pair. Seeds with an unknown arm have no row and are not assigned zero error. †: the regions case was re-executed after a zero-step host interruption; this pair is excluded under the registered rule.

cause. Explanatory prose sometimes claimed a state while its typed field remained null, and some prose was itself visually wrong. We therefore did not backfill facts from natural-language explanations.

Fixed-frame CUDA diagnostics. We retained 12 historical states covering held, near-target, released, displaced, and occluded configurations. A legacy replay reproduced all 12 original outputs exactly. The complete diagnostic campaign contained 74 real GPU generations: two saved outputs before a statistics-script failure, a 36-generation comparison of three profiles, a 24-generation current-evidence comparison, and a 12-generation isolated grip probe. The failed first attempt and all unfavorable results remain in the evidence package; these are not 74 simulator episodes or 74 independent states.

A flat facts protocol eliminated all-five-null answers on the 12 states, but introduced high-confidence false claims. One distant held cup was accepted as aligned; another state with two-pad contact was described as released; an already separated cup was described as held. More explicit current-evidence prompts withdrew some positive votes, but yielded zero effective alignment-positive votes in both tested presentations. Removing all reference images did not resolve the false-held claim on the final released example.

The isolated grip probe, using only the three current full images, produced held=true, released=false and clear=false with confidence 0.95 on all 12 states. Four states had no gripper contact in the retained physical diagnostics; a reviewed released frame visibly showed separation. This simplification changed several aspects of the request, so it is not a clean single-factor ablation. It nevertheless shows that removing reference images and placement context was insufficient for this configuration. Self-reported confidence is not calibrated physical evidence.

The practical consequence is to keep the frozen presentation comparison distinct from an unvalidated feedback loop. The expanded point/regions run reuses the same frozen verifier in both arms; it does not silently substitute one of the failed candidate verifiers. A false visual fact can be structurally valid, and a conservative unknown can still prevent useful progress. Both precision and useful coverage are necessary before enabling feedback.

7 Limitations and Evidence Boundaries

The method depends on simulator-provided identity and geometry. In particular, an asset's receiving site is stronger information than an ordinary RGB observation, even when no reward or success threshold is read. Future RGB-only or real-robot use would require a separately evaluated perception and geometry front end. Stable handles do not show that the VLA explicitly recognizes individual region IDs: the current policy presentation contains unnumbered marks and descriptive text.

The treatment changes the neighbor points, object proxy, and explanatory legend together. A positive result would support this package under the registered controls, not a claim that finer IDs alone improve accuracy. The AABB proxy is not a semantic contact anchor. Single-pixel marks at the native resolution may be difficult to use, but visual inspection alone cannot establish whether the model understood them. No explicit image-to-action correction, adaptive mesh refinement, or learned three-dimensional accuracy mechanism is implemented.

The point-focus comparator already includes the shared stage wrapper and visual cues. This experiment therefore estimates the incremental region-presentation effect within that wrapper. It does not establish the net benefit of the complete wrapper over the unwrapped native policy; that requires a separate comparison under the same current protocol.

The earlier three-seed experiment is development evidence: the region arm succeeded in two of three episodes and the point arm in one of three. The region arm's terminal root-to-site horizontal diagnostics were 7.33248, 3.63825, and 1.26576 cm. Thus 3.64 cm is the median across all three episodes, including the failure, and happens also to be one successful episode's value. It is not a mean, a maximum-

error bound, or an accuracy guarantee. These development outcomes must remain separate from the prospectively registered 12-seed paired validation (seeds 20261031–20261042, 24 episodes). The latter provides same-task, same-policy evidence on new seeds only; the present results do not establish cross-model, cross-benchmark, or broad task generalization. Official success, diagnostic position, gripper contact, and visual verifier labels are distinct quantities and should not be substituted for one another. Unknown or invalid episodes must be accounted for explicitly, and performance claims must follow paired outcomes and uncertainty rather than selected successful examples.

8 Conclusion

We implemented persistent simulator-backed entity bindings and local spatial references at the input of an unchanged frozen robot policy. The run was interrupted once by a host shutdown; one case that had recorded no environment step was re-executed after registration, and one case ended as an unknown infrastructure outcome. Under the registered accounting rule, 10 of 12 pairs are admissible: regions succeeded in 6/10 and point focus in 6/10, with 0 region-only and 0 point-only successes. The paired difference is 0.0 percentage points (conservative 95% interval $[-35.5, 35.5]$, exact paired $p = 1.0000$). Under the registered rule, the observed paired success difference is zero; the validation does not establish a region advantage or prove equivalence. Including the re-executed case, a disclosed deviation, adds one pair (seed 20261034) and gives regions 6/11 versus point focus 7/11 (-9.1 percentage points, $[-45.7, 32.7]$, $p = 1.0000$); this pair is the only source of that difference. Neither analysis supports a success benefit from the region presentation. The result concerns the combined neighborhood, object-proxy and legend treatment; it does not isolate finer symbolic IDs or establish cross-model, cross-benchmark or real-world transfer. The failed visual-feedback diagnostics further show why structured responses and high confidence cannot substitute for physical verification. The complete registered ledger, source identities and negative evidence are retained for subsequent work.

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A Reproducibility and Interpretation

Artifacts. The release retains the registered matrix, source and checkpoint asset digests, execution identities, per-episode results, raw observations, policy inputs, unmodified action chunks, planner responses, and offline analysis receipts. Earlier development sources are not overwritten. The manuscript's figures are generated from saved data rather than synthesized success scenes. Model weights are not duplicated in the result archive.

What is and is not an ID treatment. Child IDs are deterministic handles for projected reference points, used by the execution and audit machinery. The frozen action policy sees colored glyphs and the explanatory legend, not the hash strings or a numbered action menu. The study therefore cannot show that a VLA learned to understand finer symbolic IDs. To test that separate hypothesis, one would need controls for label content, temporal stability, neighborhood geometry, object proxy, and legend wording. Those ablations have not been run here.

Fixed explanatory legend.

> When visible, cyan marks the object's geometric center proxy; discrete orange neighboring reference points show the destination's local spatial context. The proxy is not an exact mouth or contact point; use the object shape and all camera views.

The sentence supplements the original placement task. It does not declare the center proxy to be the physically correct contact location or specify a world-coordinate action.

Diagnostic safeguards. Raw booleans and nulls are retained. A missing field, wrong type, invalid schema, or stale request binding is not a negative physical label and is not repaired by guessing. A visual explanation is not promoted into a typed fact. The corrected feedback prototype requires two fresh consecutive positive confirmations with the existing 0.8 confidence gate, but is not enabled in the expanded presentation comparison. The absence of training updates and the use of CUDA are verified implementation properties, not task performance measurements.

Scope of completion. A complete execution record may contain known failures. Full case accounting can also contain unknown infrastructure outcomes; it is not synonymous with 24 completed robot trajectories. Results must state both counts. Successful compilation, protocol validity, and passing unit tests likewise cannot substitute for physical task outcomes. The present version concerns one model and one task; cross-model, cross-benchmark and real-robot evaluation remain unperformed for the region treatment.

B Prospective Per-Seed Ledger

Table 3: Complete registered seed ledger. S: success; F: known failure; U: unknown. Steps are executed control steps. XY is the offline terminal diagnostic in centimeters; missing evidence is shown as –. "Pair" denotes admission for paired inference under the registered rule ("No/Yes": admitted only when the re-executed case is included). †: re-executed once after a host shutdown that preceded its first environment step (Section 4); unknown under the registered rule. ‡: steps executed before an infrastructure timeout; no official outcome or terminal diagnostic was recorded.

	Point			Regions			
Seed	Out	Steps	XY	Out	Steps	XY	Pair
20261031	F	600	16.29	F	600	17.67	Yes
20261032	F	600	11.81	F	600	7.40	Yes
20261033	F	600	19.43	F	600	19.18	Yes
20261034	S	319	3.39	F†	600	12.10	No/Yes
20261035	F	600	6.78	F	600	8.22	Yes
20261036	S	256	0.67	S	258	0.37	Yes
20261037	S	281	3.14	S	290	0.31	Yes
20261038	S	295	1.51	S	289	0.97	Yes
20261039	S	280	3.72	S	296	2.13	Yes
20261040	S	274	2.03	S	295	1.71	Yes
20261041	S	262	0.19	S	290	1.87	Yes
20261042	U	544‡	–	F	600	15.99	No

Statistical interval definitions. Per-arm intervals are Wilson 95% score intervals on the admitted paired subset. The paired risk-difference interval subtracts the endpoints of two 97.5% Clopper–Pearson intervals, one for region-only success probability and one for point-only success probability. A Bonferroni argument gives at least 95% simultaneous coverage under the paired Bernoulli sampling model; the resulting difference interval is conservative. The two-sided exact McNemar test conditions on the number of discordant pairs. The preregistered primary contrast is regions minus point.

Frozen identities. The validation manifest digest is `9414bd65...964eb9e4`. The Xiaomi checkpoint asset digest is `ccaa6517...6fc184`; the Qwen checkpoint asset digest is `6efa4249...512d6`. These are experiment asset hashes, not upstream repository commit identifiers. Full digests and file mappings are retained in the evidence package.

Seed 20261042, point. Unknown outcome (infrastructure timeout). The episode had executed 544 of its 600 steps when one planner/verifier request exceeded the 180-second client timeout (logged as `TimeoutError: timed out`). Following the registered behavior, the worker stopped; the case was not repeated, and its pair is excluded from paired inference.